

CS 350 Module Six Lab Guide

Up to this point in the course, you have worked with two input methods: the button and the serial port (UART). In this lab, you will expand your capabilities to a third input mechanism by adding a temperature and humidity sensor to your circuit. This will be the last new type of hardware you add to your circuit, although you will add two more buttons for the final project in Module Seven.

Step 1: Installing Packages

You will need to install an additional Python package to work with the new hardware. You will use the pip3 package manager to install this, like the packages you installed to support the display module.

```
sudo pip3 install adafruit-circuitpython-ahtx0
```



```
jayme — jayme@cs350lab: ~ — ssh jayme@10.67.153.71 — 96x22
[jayme@cs350lab:~]$ sudo pip3 install adafruit-circuitpython-ahtx0
[[sudo] password for jayme:
Collecting adafruit-circuitpython-ahtx0
  Downloading adafruit_circuitpython_ahtx0-1.0.21-py3-none-any.whl.metadata (3.9 kB)
Requirement already satisfied: Adafruit-Blinka in /usr/local/lib/python3.12/dist-packages (from
adafruit-circuitpython-ahtx0) (8.47.0)
Collecting adafruit-circuitpython-register (from adafruit-circuitpython-ahtx0)
  Downloading adafruit_circuitpython_register-1.10.0-py3-none-any.whl.metadata (9.3 kB)
Requirement already satisfied: adafruit-circuitpython-busdevice in /usr/local/lib/python3.12/dis
t-packages (from adafruit-circuitpython-ahtx0) (5.2.9)
Requirement already satisfied: Adafruit-PlatformDetect>=3.70.1 in /usr/local/lib/python3.12/dist
-packages (from Adafruit-Blinka->adafruit-circuitpython-ahtx0) (3.74.0)
Requirement already satisfied: Adafruit-PureIO>=1.1.7 in /usr/local/lib/python3.12/dist-packages
 (from Adafruit-Blinka->adafruit-circuitpython-ahtx0) (1.1.11)
Requirement already satisfied: binho-host-adapter>=0.1.6 in /usr/local/lib/python3.12/dist-packa
ges (from Adafruit-Blinka->adafruit-circuitpython-ahtx0) (0.1.6)
Requirement already satisfied: pyftdi>=0.40.0 in /usr/local/lib/python3.12/dist-packages (from A
dafruit-Blinka->adafruit-circuitpython-ahtx0) (0.55.4)
Requirement already satisfied: adafruit-circuitpython-typing in /usr/local/lib/python3.12/dist-p
ackages (from Adafruit-Blinka->adafruit-circuitpython-ahtx0) (1.11.0)
Requirement already satisfied: sysv-ipc>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from
Adafruit-Blinka->adafruit-circuitpython-ahtx0) (1.1.0)
```

Step 2: Power Off Your Raspberry Pi

Your Raspberry Pi must be shut down and unplugged to continue. To safely shut down the operating system running on your Raspberry Pi, enter the following command from a terminal (or the console) of the Raspberry Pi:

```
sudo halt
```

You will have to enter your password as you are executing the halt command with root privileges by calling it with *sudo*. Please wait at least two minutes after your terminal window closes before unplugging your Raspberry Pi.

Step 3: Building the Circuit

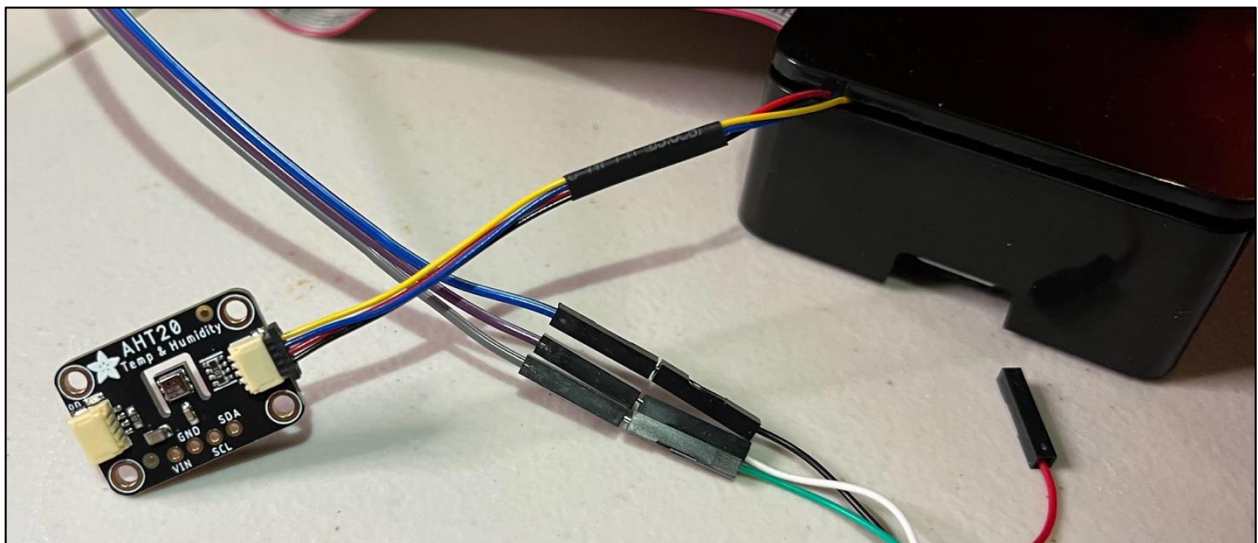
Components

- The circuit you have been dealing with in your previous labs
- The temperature sensor board



This should be an easy connection to make because you will be using the QWIIC cable that was installed on your Raspberry Pi during the initial setup. QWIIC is just a Quick Way to Connect I2C bus peripherals, and it allows you to daisy-chain multiple sensors as long as they can be set to different bus addresses. This sensor utilizes I2C address 0x38, so you could only use additional sensors that have different I2C addresses.

Connect the board to the QWIIC cable. You can use either port, as the board is set up for pass-through connections to allow additional sensors to be added to the I2C bus. The connector will only fit one way into the socket; a small bump on the connector allows it to fit properly. This should complete adding the component.



Step 4: Power Up Your Raspberry Pi

Reconnect your Raspberry Pi to power, and once it has booted, log in.

To test your sensor, you can run the `TemperatureSensorTest.py` script, which is part of the source code for this module. The script has been fully documented so that you can see what is necessary to interact with the temperature sensor. Run the script with:

```
python3 TemperatureSensorTest.py
```



```
jayme@cs350lab:~$ python3 TemperatureSensorTest.py
Temperature: 18.5 C
RH: 43.9 %

Temperature: 18.5 C
RH: 43.8 %

Temperature: 18.5 C
RH: 43.9 %

Temperature: 18.5 C
RH: 43.8 %

Temperature: 18.5 C
RH: 43.9 %

Temperature: 18.5 C
RH: 43.8 %

Temperature: 18.5 C
RH: 43.9 %
```

As a result, you should see the current temperature and humidity in your location.

Step 5: Integrating With Other Capabilities

At this point, you should have a working temperature and humidity sensor. However, the data is only being displayed on the application console. This is where you must consider how it can be integrated with the other components of your circuit.

In this step, you will edit the software template provided in the source code to meet the following requirements:

1. Displays the temperature and relative humidity on one line of your 16x2 display
2. Displays the scale of the temperature, either as Fahrenheit (F) or Celsius (C)
3. Displays the humidity as a percentage (%)
4. Displays the current date and time on the other line of your 16x2 display
5. Swaps the temperature scales between Fahrenheit and Celsius with a button push

When your application is complete, you should run it with the following command:

```
sudo python3 TemperatureSensorIntegration.py
```

Your results should look similar to the following image:

